DBMS Lab Assignment 3 - Solution

SELECT queries on the EMPLOYEE table: aliases, arithmetic, NULL handling, BETWEEN, IN, LIKE and UPDATE. Each question shows the Oracle SQL followed by its output.

*Note: outputs were produced by running equivalent statements on a SQLite database with the same table and data, formatted like SQL*Plus.*

Create table EMPLOYEE

SQL

```
CREATE TABLE EMPLOYEE (
  EMPLOYEE_ID   NUMBER(6),
  LAST_NAME     VARCHAR2(25),
  JOB_ID        VARCHAR2(10),
  SALARY        NUMBER(8,2),
  COMM_PCT      NUMBER(4,2),
  MGR_ID        NUMBER(6),
  DEPARTMENT_ID NUMBER(4));

DESC EMPLOYEE;
```

OUTPUT

```
Table created.

SQL> DESC EMPLOYEE
Name           Null?  Type
-------------  -----  ------------
EMPLOYEE_ID           NUMBER(6)
LAST_NAME             VARCHAR2(25)
JOB_ID                VARCHAR2(10)
SALARY                NUMBER(8,2)
COMM_PCT              NUMBER(4,2)
MGR_ID                NUMBER(6)
DEPARTMENT_ID         NUMBER(4)
```

Note: The sheet lists no constraints, so none are added.

Insert the data

SQL

```
INSERT INTO EMPLOYEE VALUES (198,'Connell','SH_CLERK',2600,2.5,124,50);
INSERT INTO EMPLOYEE VALUES (199,'Grant','SH_CLERK',2600,2.2,124,50);
INSERT INTO EMPLOYEE VALUES (200,'Whalen','AD_ASST',4400,1.3,101,10);
INSERT INTO EMPLOYEE VALUES (201,'Hartstein','IT_PROG',6000,NULL,100,20);
INSERT INTO EMPLOYEE VALUES (202,'Fay','AC_MGR',6500,NULL,210,20);
INSERT INTO EMPLOYEE VALUES (203,'Mavris','AD_VP',7500,NULL,101,40);
INSERT INTO EMPLOYEE VALUES (204,'Baer','AD_PRES',3500,1.5,101,90);
INSERT INTO EMPLOYEE VALUES (205,'Higgins','AC_MGR',2300,NULL,101,60);
INSERT INTO EMPLOYEE VALUES (206,'Gitz','IT_PROG',5000,NULL,103,60);
INSERT INTO EMPLOYEE VALUES (100,'King','AD_ASST',8956,0.3,108,100);
INSERT INTO EMPLOYEE VALUES (101,'Kochar','SH_CLERK',3400,1.3,118,30);
COMMIT;

SELECT * FROM EMPLOYEE;
```

OUTPUT

```
11 rows created.

EMPLOYEE_ID  LAST_NAME  JOB_ID    SALARY  COMM_PCT  MGR_ID  DEPARTMENT_ID
-----------  ---------  --------  ------  --------  ------  -------------
        198  Connell    SH_CLERK    2600       2.5     124             50
        199  Grant      SH_CLERK    2600       2.2     124             50
        200  Whalen     AD_ASST     4400       1.3     101             10
        201  Hartstein  IT_PROG     6000                100             20
        202  Fay        AC_MGR      6500                210             20
        203  Mavris     AD_VP       7500                101             40
        204  Baer       AD_PRES     3500       1.5     101             90
        205  Higgins    AC_MGR      2300                101             60
        206  Gitz       IT_PROG     5000                103             60
        100  King       AD_ASST     8956       0.3     108            100
        101  Kochar     SH_CLERK    3400       1.3     118             30

11 rows selected.
```

1. last_name, job_id, employee_id with employee_id first

SQL

```
SELECT employee_id, last_name, job_id
FROM   employee;
```

OUTPUT

```
EMPLOYEE_ID  LAST_NAME  JOB_ID
-----------  ---------  --------
        198  Connell    SH_CLERK
        199  Grant      SH_CLERK
        200  Whalen     AD_ASST
        201  Hartstein  IT_PROG
        202  Fay        AC_MGR
        203  Mavris     AD_VP
        204  Baer       AD_PRES
        205  Higgins    AC_MGR
        206  Gitz       IT_PROG
        100  King       AD_ASST
        101  Kochar     SH_CLERK

11 rows selected.
```

2. Details of all employees of department 60

SQL

```
SELECT * FROM employee WHERE department_id = 60;
```

OUTPUT

```
EMPLOYEE_ID  LAST_NAME  JOB_ID   SALARY  COMM_PCT  MGR_ID  DEPARTMENT_ID
-----------  ---------  -------  ------  --------  ------  -------------
        205  Higgins    AC_MGR     2300               101             60
        206  Gitz       IT_PROG    5000               103             60

2 rows selected.
```

3. Details of the employee whose last_name is King

SQL

```
SELECT * FROM employee WHERE last_name = 'King';
```

OUTPUT

```
EMPLOYEE_ID  LAST_NAME  JOB_ID   SALARY  COMM_PCT  MGR_ID  DEPARTMENT_ID
-----------  ---------  -------  ------  --------  ------  -------------
        100  King       AD_ASST    8956       0.3     108            100

1 row selected.
```

4. Unique job_id, with the column aliased as JOB_TITLE

SQL

```
SELECT DISTINCT job_id AS job_title
FROM   employee;
```

OUTPUT

```
JOB_TITLE
---------
AC_MGR
AD_ASST
AD_PRES
AD_VP
IT_PROG
SH_CLERK

6 rows selected.
```

Note: DISTINCT removes the repeats (SH_CLERK, IT_PROG, AD_ASST and AC_MGR each occur more than once). Oracle does not guarantee row order here.

5. last_name, salary and salary + Rs 300 as 'Increased Salary'

SQL

```
SELECT last_name, salary, salary + 300 AS "Increased Salary"
FROM   employee;
```

OUTPUT

```
LAST_NAME  SALARY  Increased Salary
---------  ------  ----------------
Connell      2600              2900
Grant        2600              2900
Whalen       4400              4700
Hartstein    6000              6300
Fay          6500              6800
Mavris       7500              7800
Baer         3500              3800
Higgins      2300              2600
Gitz         5000              5300
King         8956              9256
Kochar       3400              3700

11 rows selected.
```

6. last_name, salary and annual compensation (plus a one-time bonus of Rs 100)

SQL

```
SELECT last_name, salary,
       salary * 12 + 100 AS annual_compensation
FROM   employee;
```

OUTPUT

```
LAST_NAME  SALARY  ANNUAL_COMPENSATION
---------  ------  ------------------
Connell      2600               31300
Grant        2600               31300
Whalen       4400               52900
Hartstein    6000               72100
Fay          6500               78100
Mavris       7500               90100
Baer         3500               42100
Higgins      2300               27700
Gitz         5000               60100
King         8956              107572
Kochar       3400               40900

11 rows selected.
```

Note: Annual compensation = 12 x monthly salary, plus the one-time Rs 100 bonus added once (not 12 times).

7. Employees who get commission

SQL

```
SELECT * FROM employee WHERE comm_pct IS NOT NULL;
```

OUTPUT

```
EMPLOYEE_ID  LAST_NAME  JOB_ID    SALARY  COMM_PCT  MGR_ID  DEPARTMENT_ID
-----------  ---------  --------  ------  --------  ------  -------------
        198  Connell    SH_CLERK    2600       2.5     124             50
        199  Grant      SH_CLERK    2600       2.2     124             50
        200  Whalen     AD_ASST     4400       1.3     101             10
        204  Baer       AD_PRES     3500       1.5     101             90
        100  King       AD_ASST     8956       0.3     108            100
        101  Kochar     SH_CLERK    3400       1.3     118             30

6 rows selected.
```

8. Employees who do not get commission

SQL

```
SELECT * FROM employee WHERE comm_pct IS NULL;
```

OUTPUT

```
EMPLOYEE_ID  LAST_NAME  JOB_ID   SALARY  COMM_PCT  MGR_ID  DEPARTMENT_ID
-----------  ---------  -------  ------  --------  ------  -------------
        201  Hartstein  IT_PROG    6000               100             20
        202  Fay        AC_MGR     6500               210             20
        203  Mavris     AD_VP      7500               101             40
        205  Higgins    AC_MGR     2300               101             60
        206  Gitz       IT_PROG    5000               103             60

5 rows selected.
```

Note: IS NULL is required; comm_pct = NULL never matches.

9. employee_id, department_id, salary where salary > 5000

SQL

```
SELECT employee_id, department_id, salary
FROM   employee
WHERE  salary > 5000;
```

OUTPUT

```
EMPLOYEE_ID  DEPARTMENT_ID  SALARY
-----------  -------------  ------
        201             20    6000
        202             20    6500
        203             40    7500
        100            100    8956

4 rows selected.
```

10. last_name and salary where salary is between 4000 and 7000

SQL

```
SELECT last_name, salary
FROM   employee
WHERE  salary BETWEEN 4000 AND 7000;
```

OUTPUT

```
LAST_NAME  SALARY
---------  ------
Whalen       4400
Hartstein    6000
Fay          6500
Gitz         5000

4 rows selected.
```

Note: BETWEEN includes both limits.

11. Employees whose salary is 6000, 6500 or 7000

SQL

```
SELECT * FROM employee WHERE salary IN (6000, 6500, 7000);
```

OUTPUT

```
EMPLOYEE_ID  LAST_NAME  JOB_ID   SALARY  COMM_PCT  MGR_ID  DEPARTMENT_ID
-----------  ---------  -------  ------  --------  ------  -------------
        201  Hartstein  IT_PROG    6000               100             20
        202  Fay        AC_MGR     6500               210             20

2 rows selected.
```

12. Employees working in department 10, 20, 30 or 50

SQL

```
SELECT * FROM employee
WHERE  department_id IN (10, 20, 30, 50);
```

OUTPUT

```
EMPLOYEE_ID  LAST_NAME  JOB_ID    SALARY  COMM_PCT  MGR_ID  DEPARTMENT_ID
-----------  ---------  --------  ------  --------  ------  -------------
        198  Connell    SH_CLERK    2600       2.5     124             50
        199  Grant      SH_CLERK    2600       2.2     124             50
        200  Whalen     AD_ASST     4400       1.3     101             10
        201  Hartstein  IT_PROG     6000               100             20
        202  Fay        AC_MGR      6500               210             20
        101  Kochar     SH_CLERK    3400       1.3     118             30

6 rows selected.
```

13. Employees whose salary is not equal to 5000

SQL

```
SELECT * FROM employee WHERE salary <> 5000;
```

OUTPUT

```
EMPLOYEE_ID  LAST_NAME  JOB_ID    SALARY  COMM_PCT  MGR_ID  DEPARTMENT_ID
-----------  ---------  --------  ------  --------  ------  -------------
        198  Connell    SH_CLERK    2600       2.5     124             50
        199  Grant      SH_CLERK    2600       2.2     124             50
        200  Whalen     AD_ASST     4400       1.3     101             10
        201  Hartstein  IT_PROG     6000               100             20
        202  Fay        AC_MGR      6500               210             20
        203  Mavris     AD_VP       7500               101             40
        204  Baer       AD_PRES     3500       1.5     101             90
        205  Higgins    AC_MGR      2300               101             60
        100  King       AD_ASST     8956       0.3     108            100
        101  Kochar     SH_CLERK    3400       1.3     118             30

10 rows selected.
```

14. All the clerks

SQL

```
SELECT * FROM employee WHERE job_id LIKE '%CLERK';
```

OUTPUT

```
EMPLOYEE_ID  LAST_NAME  JOB_ID    SALARY  COMM_PCT  MGR_ID  DEPARTMENT_ID
-----------  ---------  --------  ------  --------  ------  -------------
        198  Connell    SH_CLERK    2600       2.5     124             50
        199  Grant      SH_CLERK    2600       2.2     124             50
        101  Kochar     SH_CLERK    3400       1.3     118             30

3 rows selected.
```

15. Update job_id to 'Grade_A' for employees earning more than 5000

SQL

```
UPDATE employee
SET    job_id = 'Grade_A'
WHERE  salary > 5000;
COMMIT;

SELECT * FROM employee;
```

OUTPUT

```
4 rows updated.

EMPLOYEE_ID  LAST_NAME  JOB_ID    SALARY  COMM_PCT  MGR_ID  DEPARTMENT_ID
-----------  ---------  --------  ------  --------  ------  -------------
        198  Connell    SH_CLERK    2600       2.5     124             50
        199  Grant      SH_CLERK    2600       2.2     124             50
        200  Whalen     AD_ASST     4400       1.3     101             10
        201  Hartstein  Grade_A     6000               100             20
        202  Fay        Grade_A     6500               210             20
        203  Mavris     Grade_A     7500               101             40
        204  Baer       AD_PRES     3500       1.5     101             90
        205  Higgins    AC_MGR      2300               101             60
        206  Gitz       IT_PROG     5000               103             60
        100  King       Grade_A     8956       0.3     108            100
        101  Kochar     SH_CLERK    3400       1.3     118             30

11 rows selected.
```

Note: Four employees earn more than 5000 (Gitz earns exactly 5000, so he is unchanged). From here on, the table shows these new job_ids.

16. Employees who are CLERK, PROGRAMMER or ASSISTANT

SQL

```
SELECT * FROM employee
WHERE  job_id IN ('SH_CLERK', 'IT_PROG', 'AD_ASST');
```

OUTPUT

```
EMPLOYEE_ID  LAST_NAME  JOB_ID    SALARY  COMM_PCT  MGR_ID  DEPARTMENT_ID
-----------  ---------  --------  ------  --------  ------  -------------
        198  Connell    SH_CLERK    2600       2.5     124             50
        199  Grant      SH_CLERK    2600       2.2     124             50
        200  Whalen     AD_ASST     4400       1.3     101             10
        206  Gitz       IT_PROG     5000               103             60
        101  Kochar     SH_CLERK    3400       1.3     118             30

5 rows selected.
```

Note: This runs after question 15, so Hartstein (IT_PROG) and King (AD_ASST) no longer match because their job_id is now Grade_A.

17. Employees who are CLERK with salary below 3000

SQL

```
SELECT * FROM employee
WHERE  job_id LIKE '%CLERK' AND salary < 3000;
```

OUTPUT

```
EMPLOYEE_ID  LAST_NAME  JOB_ID    SALARY  COMM_PCT  MGR_ID  DEPARTMENT_ID
-----------  ---------  --------  ------  --------  ------  -------------
        198  Connell    SH_CLERK    2600       2.5     124             50
        199  Grant      SH_CLERK    2600       2.2     124             50

2 rows selected.
```

18. last_name, mgr_id where salary is above 3000 and the manager is 101

SQL

```
SELECT last_name, mgr_id
FROM   employee
WHERE  salary > 3000 AND mgr_id = 101;
```

OUTPUT

```
LAST_NAME  MGR_ID
---------  ------
Whalen        101
Mavris        101
Baer          101

3 rows selected.
```